



Course title: Electronic Engineering

Course Code: CSE 2109



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JFET

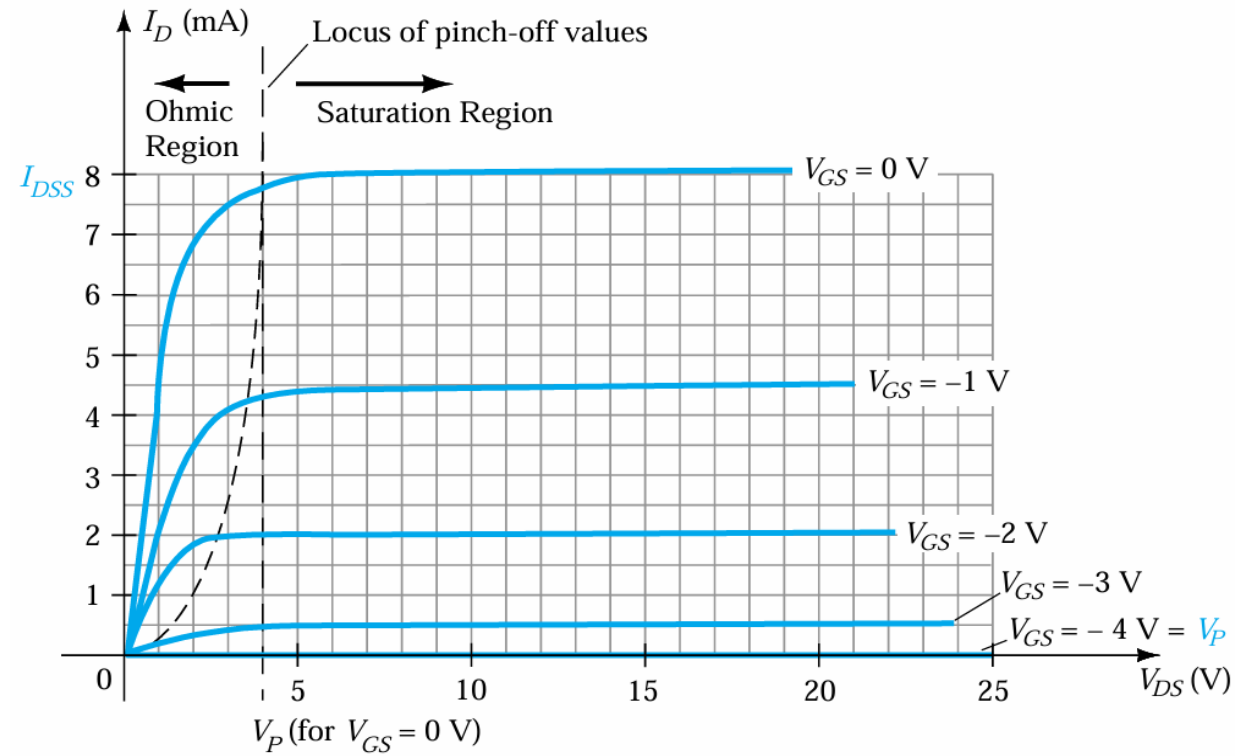


Figure 5.10 n -Channel JFET characteristics with $I_{DSS} = 8$ mA and $V_P = -4$ V.

MOSFET

By mode of operation

- Enhancement-mode MOSFET
 - Normally OFF at $V_{GS}=0$
- Depletion-mode MOSFET
 - Normally ON at $V_{GS}=0$

By channel type

- N-channel MOSFET (NMOS)
- P-channel MOSFET (PMOS)

What is the difference between Enhancement mode MOSFET and Depletion Mode MOSFET??

MOSFET

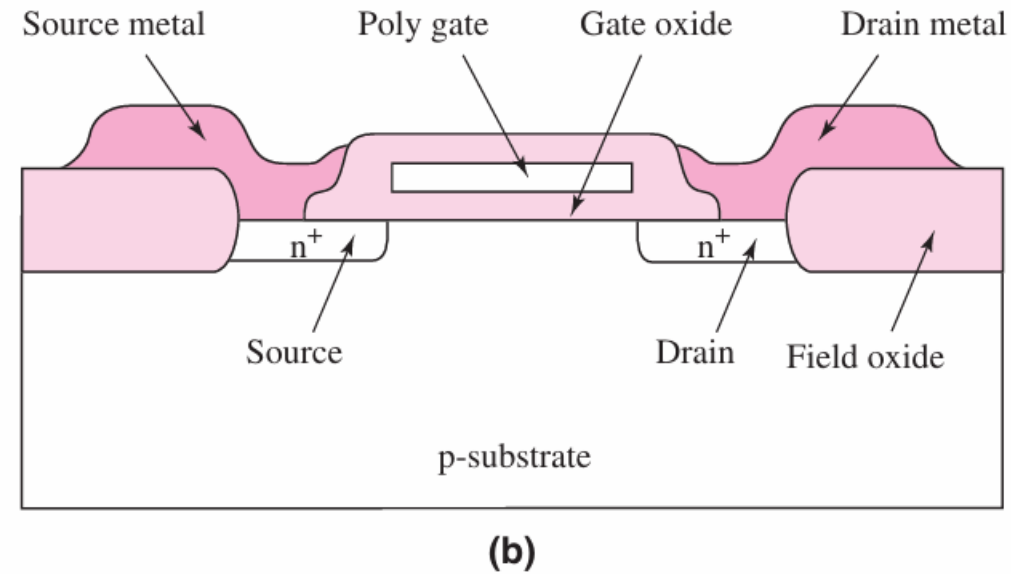
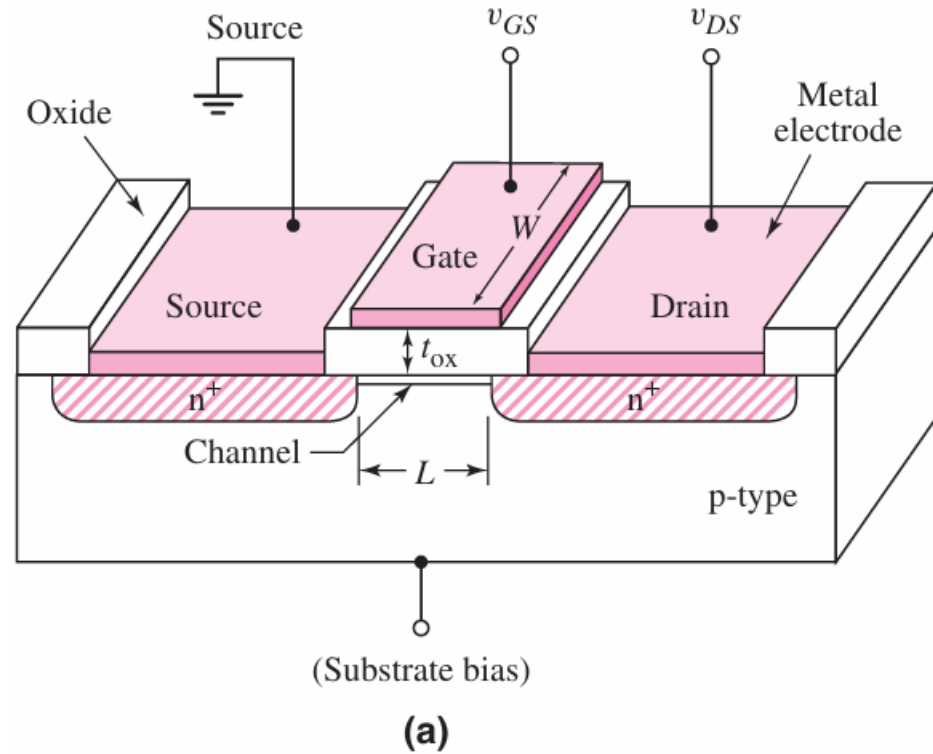


Figure 3.5 (a) Schematic diagram of an n-channel enhancement-mode MOSFET and (b) an n-channel MOSFET, showing the field oxide and polysilicon gate

MOSFET

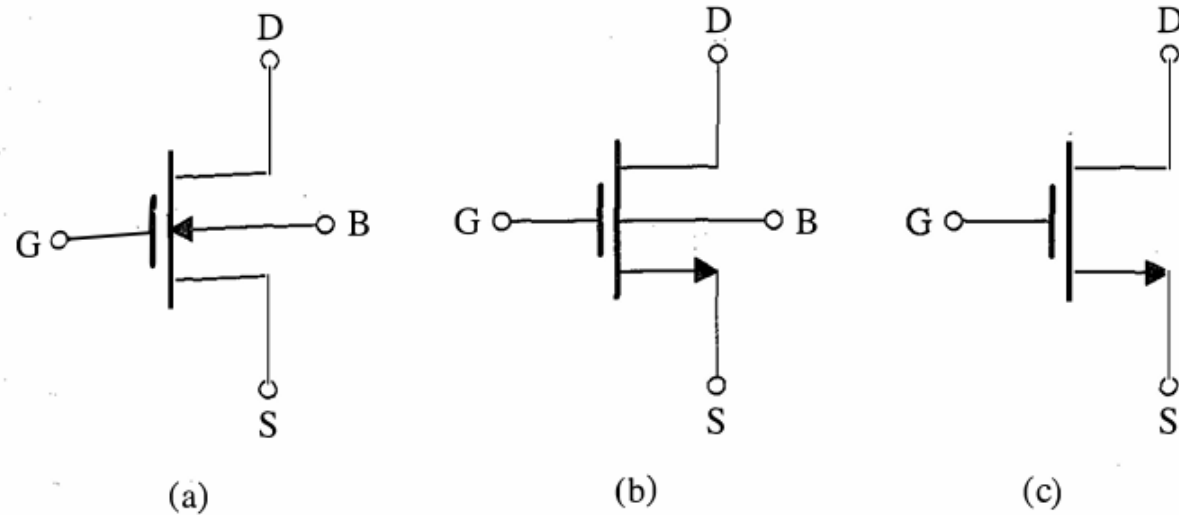
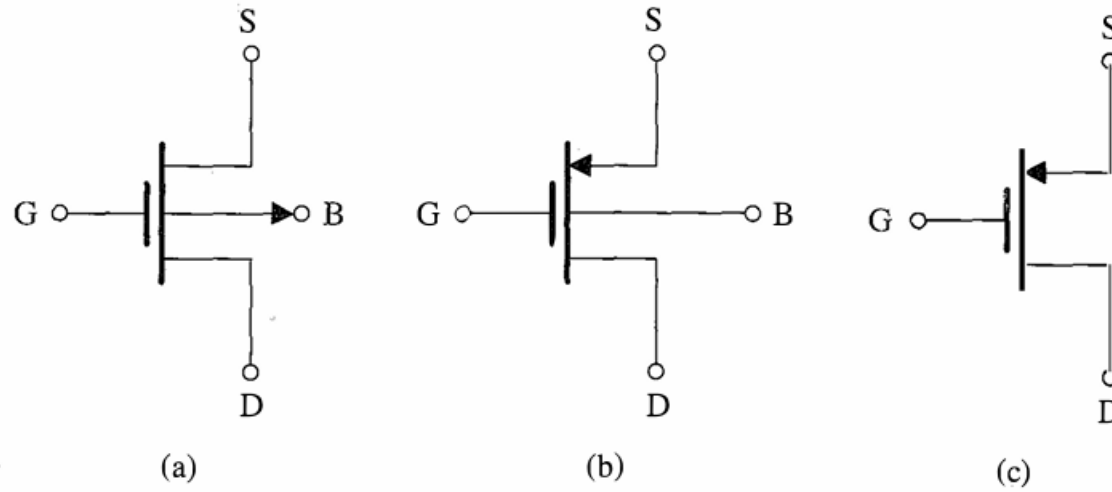


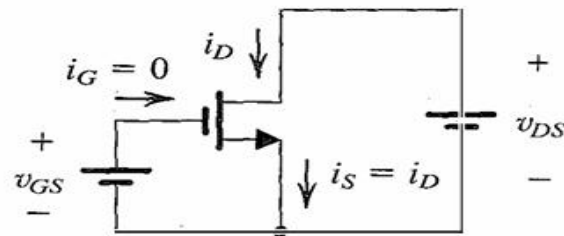
FIGURE 4.10 (a) Circuit symbol for the n -channel enhancement-type MOSFET. (b) Modified circuit symbol with an arrowhead on the source terminal to distinguish it from the drain and to indicate device polarity (i.e., n channel). (c) Simplified circuit symbol to be used when the source is connected to the body or when the effect of the body on device operation is unimportant.

MOSFET

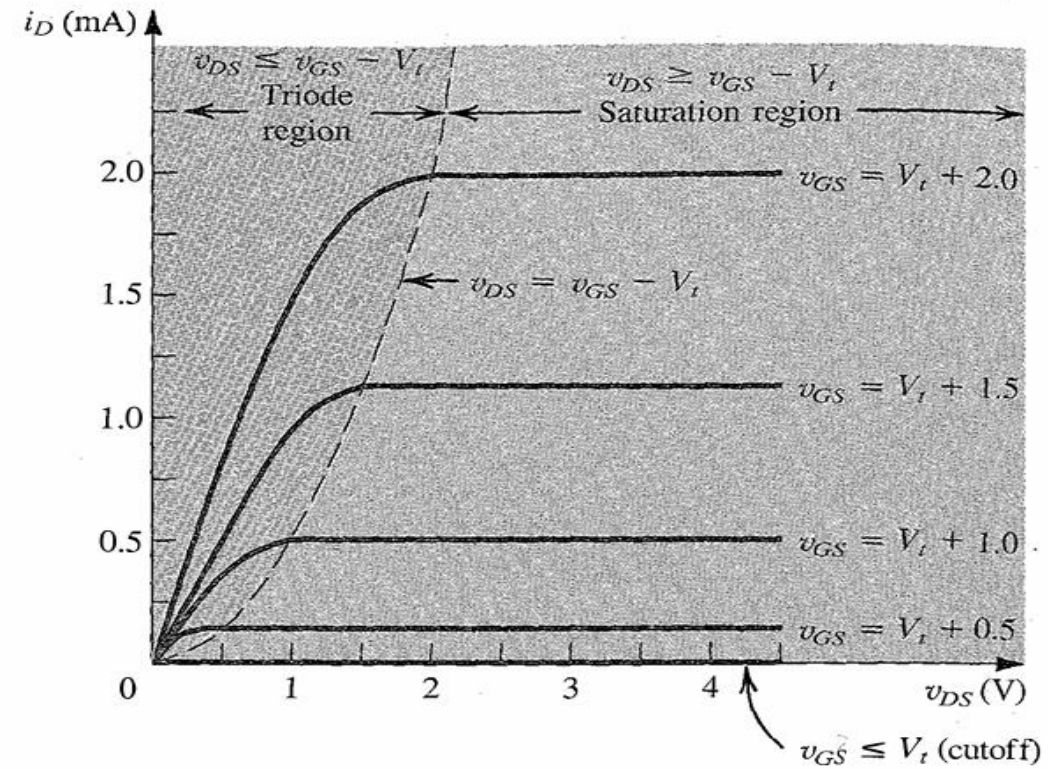


Symbol for PMOS

MOSFET



(a)



(b)

FIGURE 4.11 (a) An n -channel enhancement-type MOSFET with v_{GS} and v_{DS} applied and with the normal directions of current flow indicated. (b) The i_D - v_{DS} characteristics for a device with $k'_n (W/L) = 1.0 \text{ mA/V}^2$.

MOSFET

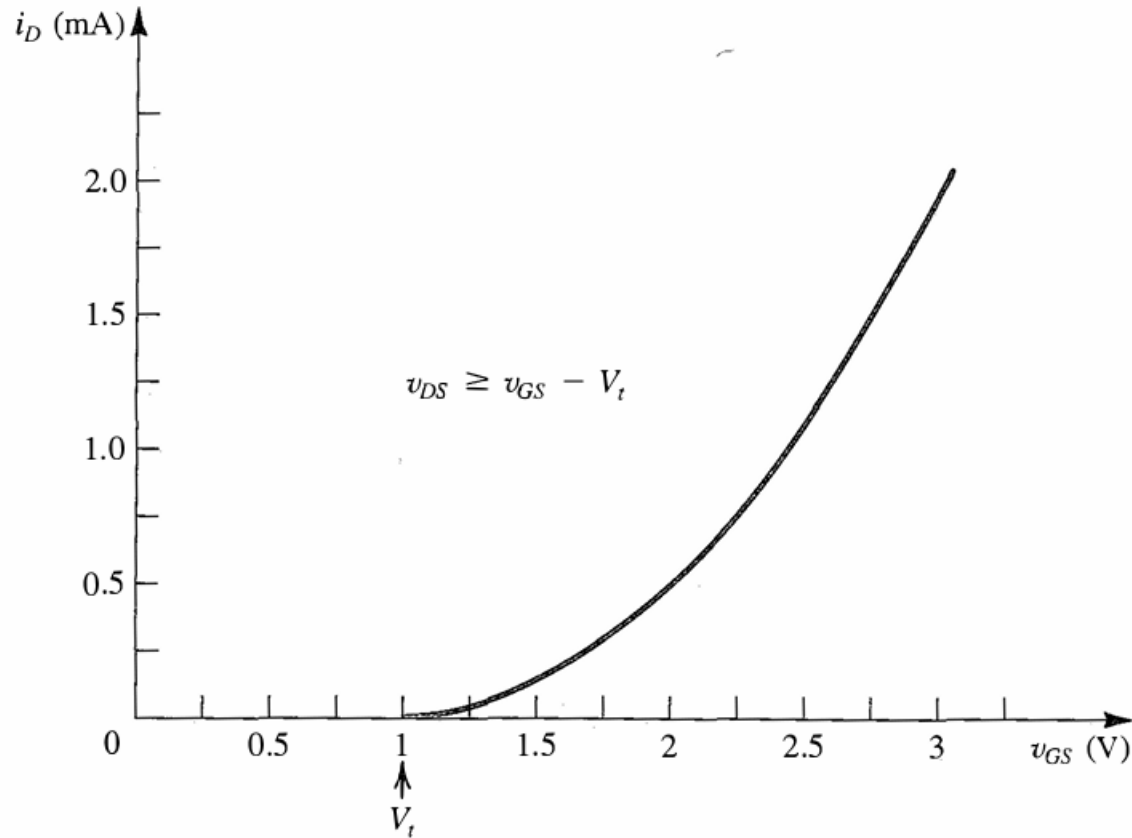
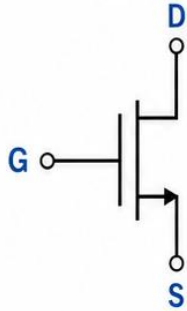


FIGURE 4.12 The i_D - v_{GS} characteristic for an enhancement-type NMOS transistor in saturation ($V_t = 1$ V, $k'_n W/L = 1.0$ mA/V²).

MOSFET

NMOS (Enhancement Type)



1 Triode / Linear Region

Condition: $V_{GS} > V_{TN}$ and $0 \leq V_{DS} < V_{GS} - V_{TN}$

$$I_D = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(V_{GS} - V_{TN}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

2 Saturation Region

Condition: $V_{GS} > V_{TN}$ and $V_{DS} \geq V_{GS} - V_{TN}$

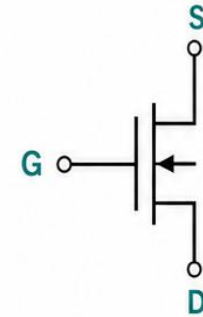
$$I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_{TN})^2$$

i Overdrive voltage: $V_{OV} = V_{GS} - V_{TN}$

Symbols

- μ_n, μ_p = electron and hole mobility
- C_{ox} = oxide capacitance per unit area
- $\frac{W}{L}$ = transistor width-to-length ratio
- V_{TN} = NMOS threshold voltage
- V_{TP} = PMOS threshold voltage
- I_D = drain current

PMOS (Enhancement Type)



1 Triode / Linear Region

Condition: $V_{SG} > |V_{TP}|$ and $0 \leq V_{SD} < V_{SG} - |V_{TP}|$

$$|I_D| = \mu_p C_{ox} \left(\frac{W}{L} \right) \left[(V_{SG} - |V_{TP}|) V_{SD} - \frac{V_{SD}^2}{2} \right]$$

2 Saturation Region

Condition: $V_{SG} > |V_{TP}|$ and $V_{SD} \geq V_{SG} - |V_{TP}|$

$$|I_D| = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right) (V_{SG} - |V_{TP}|)^2$$

i Overdrive voltage: $V_{OV} = V_{SG} - |V_{TP}|$

MOSFET

Objective: Calculate the current in an n-channel MOSFET.

Consider an n-channel enhancement-mode MOSFET with the following parameters: $V_{TN} = 0.4 \text{ V}$, $W = 20 \mu\text{m}$, $L = 0.8 \mu\text{m}$, $\mu_n = 650 \text{ cm}^2/\text{V-s}$, $t_{\text{ox}} = 200 \text{ \AA}$, and $\epsilon_{\text{ox}} = (3.9)(8.85 \times 10^{-14}) \text{ F/cm}$. Determine the current when the transistor is biased in the saturation region for (a) $v_{GS} = 0.8 \text{ V}$ and (b) $v_{GS} = 1.6 \text{ V}$.

Solution: The conduction parameter is determined by Equation (3.3(a)). First, consider the units involved in this equation, as follows:

$$K_n = \frac{W(\text{cm}) \cdot \mu_n \left(\frac{\text{cm}^2}{\text{V-s}} \right) \epsilon_{\text{ox}} \left(\frac{\text{F}}{\text{cm}} \right)}{2L(\text{cm}) \cdot t_{\text{ox}}(\text{cm})} = \frac{\text{F}}{\text{V-s}} = \frac{(\text{C/V})}{\text{V-s}} = \frac{\text{A}}{\text{V}^2}$$

The value of the conduction parameter is therefore

$$K_n = \frac{W \mu_n \epsilon_{\text{ox}}}{2L t_{\text{ox}}} = \frac{(20 \times 10^{-4})(650)(3.9)(8.85 \times 10^{-14})}{2(0.8 \times 10^{-4})(200 \times 10^{-8})}$$

or

$$K_n = 1.40 \text{ mA/V}^2$$

From Equation (3.2(b)), we find:

(a) For $v_{GS} = 0.8 \text{ V}$,

$$i_D = K_n(v_{GS} - V_{TN})^2 = (1.40)(0.8 - 0.4)^2 = 0.224 \text{ mA}$$

(b) For $v_{GS} = 1.6 \text{ V}$,

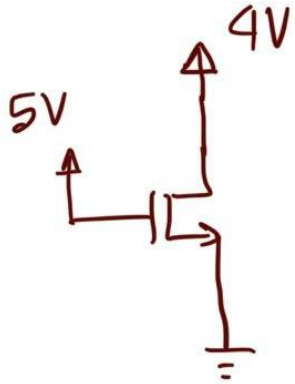
$$i_D = (1.40)(1.6 - 0.4)^2 = 2.02 \text{ mA}$$

MOSFET

Ex 3.1: An NMOS transistor with $V_{TN} = 1$ V has a drain current $i_D = 0.8$ mA when $v_{GS} = 3$ V and $v_{DS} = 4.5$ V. Calculate the drain current when: (a) $v_{GS} = 2$ V, $v_{DS} = 4.5$ V; and (b) $v_{GS} = 3$ V, $v_{DS} = 1$ V. (Ans. (a) 0.2 mA (b) 0.6 mA)

MOSFET

(*)



$$V_{Th} = 2V$$

$$k'_n = 1 \text{ mA/V}^2$$

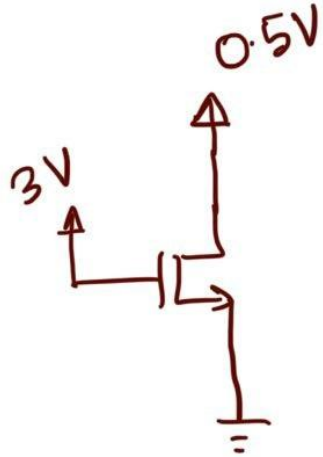
↓
process transconductance
parameter

$$\frac{W}{L} = 4$$

$$I_D = ?$$

MOSFET

(*)



$$V_{Th} = 1V$$

$$K'_n = 2 \text{ mA/V}^2$$

↓
process transconductance
parameter

$$\frac{W}{L} = 2$$

$$I_D = ?$$